11

Thermodynamic Relations, Equilibrium and Stability

Learning Objectives

After reading this chapter you should be able to

LO 11.1 Explain important mathematical theorems to obtain exact differential relation among thermodynamic variables

LO 11.2 Interpret Maxwell's four equations utilizing concept of exact differential

LO 11.3 Explain TdS first and second equations and its important deductions

LO 11.4 Discuss the impact of heat capacities ratio during the compression work

LO 11.5 Describe energy equation and its applications

LO 11.6 Explain Joule-Kelvin effect and the significance of inversion curve

LO 11.7 Formulate Clausius-Clapeyron equation and discuss how it affects phase change

LO 11.8 Evaluate thermodynamic properties other than P, V and T from equation of state

LO 11.9 Determine some important deductions based on equation of state

LO 11.10 Discuss the system comprising mixtures of variable composition

LO 11.11 Deduce the condition of thermodynamic equilibrium in a heterogeneous system

LO 11.12 Describe the Gibbs phase rule applicable for a non-reactive heterogeneous system

LO 11.13 Discuss the establishment of four types of equilibrium and their attainment

11.1 SOME MATHEMATICAL THEOREMS

LO 11.1
Explain important mathematical theorems to obtain exact differential relation among thermodynamic variables

Theorem 1 If a relation exists among the variables x, y and z, then z may be expressed as a function of x and y, or

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

If
$$\left(\frac{\partial z}{\partial x}\right)_y = M, \quad \text{and} \quad \left(\frac{\partial z}{\partial y}\right)_x = N$$

then
$$dz = M\,dx + N\,dy,$$

where z, M and N are functions of x and y. Differentiating M partially with respect to y, and N with respect to x

$$\left(\frac{\partial M}{\partial y}\right)_x = \frac{\partial^2 z}{\partial x\,\partial y}$$

$$\left(\frac{\partial N}{\partial x}\right)_y = \frac{\partial^2 z}{\partial y\,\partial x}$$

$$\therefore \quad \left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y \tag{11.1}$$

This is the *condition of exact* (*or perfect*) *differential.*

Theorem 2 If a quantity f is a function of x, y and z, and a relation exists among x, y and z, then f is a function of any two of x, y and z. Similarly, any one of x, y and z may be regarded to be a function of f and any one of x, y and z. Thus, if

$$x = x\,(f, y)$$

$$dx = \left(\frac{\partial x}{\partial f}\right)_y df + \left(\frac{\partial x}{\partial y}\right)_f dy$$

Similarly, if

$$y = y(f, z)$$

$$dy = \left(\frac{\partial y}{\partial f}\right)_z df + \left(\frac{\partial y}{\partial z}\right)_f dz$$

Substituting the expression of dy in the preceding equation

$$dx = \left(\frac{\partial x}{\partial f}\right)_y df + \left(\frac{\partial x}{\partial y}\right)_f \left[\left(\frac{\partial y}{\partial f}\right)_z df + \left(\frac{\partial y}{\partial z}\right)_f dz\right]$$

$$= \left[\left(\frac{\partial x}{\partial f}\right)_y + \left(\frac{\partial x}{\partial y}\right)_f \left(\frac{\partial y}{\partial f}\right)_z\right] df + \left(\frac{\partial x}{\partial y}\right)_f \left(\frac{\partial y}{\partial z}\right)_f dz$$

Again

$$dx = \left(\frac{\partial x}{\partial f}\right)_z df + \left(\frac{\partial x}{\partial z}\right)_f dz$$

$$\therefore \qquad \left(\frac{\partial x}{\partial z}\right)_f = \left(\frac{\partial x}{\partial y}\right)_f \left(\frac{\partial y}{\partial z}\right)_f$$

$$\therefore \qquad \boxed{\left(\frac{\partial x}{\partial y}\right)_f \left(\frac{\partial y}{\partial z}\right)_f \left(\frac{\partial z}{\partial x}\right)_f = 1} \qquad (11.2)$$

Theorem 3 Among the variables x, y and z, any one variable may be considered as a function of the other two. Thus,

$$x = x(y, z)$$

$$dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y dz$$

Similarly,

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

$$\therefore \qquad dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y \left[\left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy\right]$$

$$= \left[\left(\frac{\partial x}{\partial y}\right)_z + \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x\right] dy + \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial x}\right)_y dx$$

$$= \left[\left(\frac{\partial x}{\partial y}\right)_z + \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x\right] dy + dx$$

$$\therefore \qquad \left(\frac{\partial x}{\partial y}\right)_z + \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x = 0$$

$$\text{or} \qquad \boxed{\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial z}{\partial x}\right)_y \left(\frac{\partial y}{\partial z}\right)_x = -1} \qquad (11.3)$$

Among the thermodynamic variables p, V and T, the following relation holds good:

$$\left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V = -1$$

11.2 MAXWELL'S EQUATIONS

LO 11.2 Interpret Maxwell's four equations utilizing concept of exact differential

A pure substance existing in a single phase has only two independent variables. Of the eight quantities p, V, T, S, U, H, F (Helmholtz function) and G (Gibbs function) *any one may be expressed as a function of any two others.*

For a pure substance undergoing an infinitesimal reversible process

1. $dU = TdS - pdV$
2. $dH = dU + pdV + Vdp = TdS + Vdp$
3. $dF = dU - TdS - SdT = -pdV - SdT$
4. $dG = dH - TdS - SdT = Vdp - SdT$

Since U, H, F and G are thermodynamic properties and exact differentials of the type

$$dz = M\,dx + N\,dy, \text{ then}$$

$$\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$$

Applying this to the four equations

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V \tag{11.4}$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p \tag{11.5}$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T \tag{11.6}$$

$$\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T \tag{11.7}$$

These four equations are known as *Maxwell's equations.*

11.3 *TdS* EQUATIONS

LO 11.3
Explain *TdS* first and second equations and its important deductions

Let entropy S be imagined as a function of T and V. Then

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV$$

$$\therefore \quad TdS = T\left(\frac{\partial S}{\partial T}\right)_V dT + T\left(\frac{\partial S}{\partial V}\right)_T dV$$

Since $T\left(\frac{\partial S}{\partial T}\right)_V = C_v$, heat capacity at constant volume, and

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V, \text{ Maxwell's third equation,}$$

$$\boxed{TdS = C_v\,dT + T\left(\frac{\partial p}{\partial T}\right)_V dV} \tag{11.8}$$

This is known as the *first TdS equation.*

If $S = S(T, p)$

$$dS = \left(\frac{\partial S}{\partial T}\right)_p dT + \left(\frac{\partial S}{\partial p}\right)_T dp$$

$$\therefore \qquad TdS = T\left(\frac{\partial S}{\partial T}\right)_p dT + T\left(\frac{\partial S}{\partial p}\right)_T dp$$

Since $\quad T\left(\frac{\partial S}{\partial T}\right)_p = C_p$, and $\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$

then

$$\boxed{TdS = C_p\, dT - T\left(\frac{\partial V}{\partial T}\right)_p dp} \tag{11.9}$$

This is known as the *second TdS equation.*

11.4 DIFFERENCE IN HEAT CAPACITIES

Equating the first and second *TdS* equations

$$TdS = C_p\, dT - T\left(\frac{\partial V}{\partial T}\right)_p dp = C_v\, dT + T\left(\frac{\partial p}{\partial T}\right)_V dV$$

$$(C_p - C_v)\, dT = T\left(\frac{\partial p}{\partial T}\right)_V dV + T\left(\frac{\partial V}{\partial T}\right)_p dp$$

$$\therefore \qquad dT = \frac{T\left(\frac{\partial p}{\partial T}\right)_V}{C_p - C_v} dV + \frac{T\left(\frac{\partial V}{\partial T}\right)_p}{C_p - C_v} dp$$

Again $\quad dT = \left(\frac{\partial T}{\partial V}\right)_p dV + \left(\frac{\partial T}{\partial p}\right)_v dp$

$$\therefore \qquad \frac{T\left(\frac{\partial p}{\partial T}\right)_V}{C_p - C_v} = \left(\frac{\partial T}{\partial V}\right)_p \quad \text{and} \quad \frac{T\left(\frac{\partial V}{\partial T}\right)_p}{C_p - C_v} = \left(\frac{\partial T}{\partial p}\right)_V$$

Both these equations give

$$C_p - C_v = T\left(\frac{\partial p}{\partial T}\right)_v \left(\frac{\partial V}{\partial T}\right)_p$$

But $\quad \left(\frac{\partial p}{\partial T}\right)_v \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1$

$$\therefore \qquad \boxed{C_p - C_v = -T\left(\frac{\partial V}{\partial T}\right)_T^2 \left(\frac{\partial p}{\partial V}\right)_T} \tag{11.10}$$

This is a very important equation in thermodynamics. It indicates the following important facts:

1. Since $\left(\frac{\partial V}{\partial T}\right)_p^2$ is always positive, and $\left(\frac{\partial p}{\partial V}\right)_T$ for any substance is negative, $(C_p - C_v)$ is always positive. Therefore, C_p is always greater than C_v.
2. As $T \to 0\ K$, $C_p \to C_v$ or at absolute zero, $C_p = C_v$.
3. When $\left(\frac{\partial V}{\partial T}\right)_p = 0$ (e.g. for water at 4°C, when density is maximum, or specific volume minimum), $C_p = C_v$.
4. For an ideal gas, $pV = mRT$

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{mR}{p} = \frac{V}{T}$$

and
$$\left(\frac{\partial p}{\partial V}\right)_T = -\frac{mRT}{V^2}$$

∴
$$C_p - C_v = mR$$

or
$$\boxed{c_p - c_v = R}$$

Equation (11.10) may also be expressed in terms of *volume expansivity* (β), defined as

$$\beta = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$$

and *isothermal compressibility* (k_T), defined as

$$k_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$$

$$C_p - C_v = \frac{TV\left[\frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p\right]^2}{-\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T}$$

∴
$$C_p - C_v = \frac{TV\beta^2}{k_T} \tag{11.11}$$

LO 11.4 Discuss the impact of heat capacities ratio during the compression work

11.5 RATIO OF HEAT CAPACITIES

At constant S, the two TdS equations become

$$C_p\, dT_s = T\left(\frac{\partial V}{\partial T}\right)_p dp_s$$

$$C_v\,dT_s = -T\left(\frac{\partial p}{\partial T}\right)_V dV_s$$

$$\therefore \qquad \frac{C_p}{C_v} = -\left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V \left(\frac{\partial p}{\partial V}\right)_S = \frac{\left(\frac{\partial p}{\partial V}\right)_S}{\left(\frac{\partial p}{\partial V}\right)_T} = \gamma$$

Since $\gamma > 1$,

$$\left(\frac{\partial p}{\partial V}\right)_S > \left(\frac{\partial p}{\partial V}\right)_T$$

Therefore, the slope of an isentrope is greater than that of an isotherm on $p - v$ diagram (Fig. 11.1). For reversible and adiabatic compression, the work done is

$$W_s = h_{2s} - h_1 = \int_1^{2S} v\,dp$$

$$= \text{Area } 1 - 2S - 3 - 4 - 1$$

For reversible and isothermal compression, the work done would be

$$W_T = h_{2T} - h_1 = \int_1^{2T} v\,dp$$

$$= \text{Area } 1\text{–}2T - 3 - 4 - 1$$

$$\therefore \qquad W_T < W_S$$

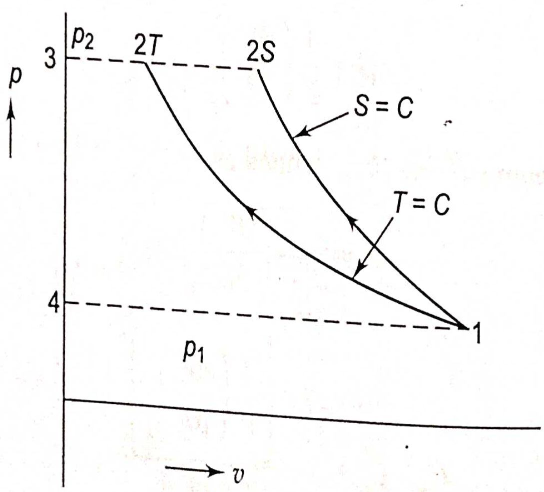

Fig. 11.1 *Compression work in different reversible process*

For polytropic compression with $1 < n < \gamma$, the work done will be between these two values. So, isothermal compression requires minimum work and isentropic compression requires the maximum work.

The adiabatic compressibility (k_s) is defined as

$$k_s = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_s$$

$$\therefore \quad \frac{C_p}{C_v} = \frac{-\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T}{-\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_s} = \gamma$$

or

$$\gamma = \frac{k}{k_s} \tag{11.12}$$

11.6 ENERGY EQUATION

LO 11.5 Describe energy equation and its applications

For a system undergoing an infinitesimal reversible process between two equilibrium states, the change of internal energy is

$$dU = TdS - pdV$$

Substituting the first TdS equation

$$dU = C_v\,dT + T\left(\frac{\partial p}{\partial T}\right)_V dV - pdV$$

$$= C_v\,dT + \left[T\left(\frac{\partial p}{\partial T}\right)_V - p\right] dV \tag{11.13}$$

If $\quad U = U(T, V)$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p \tag{11.14}$$

This is known as the *energy equation*. Two applications of the equation are given below:

1. For an ideal gas, $\quad p = \dfrac{n\bar{R}T}{V}$

$$\therefore \quad \left(\frac{\partial p}{\partial T}\right)_V = \frac{n\bar{R}}{V} = \frac{p}{T}$$

$$\therefore \quad \left(\frac{\partial U}{\partial V}\right)_T = T\cdot\frac{p}{T} - p = 0$$

U does not change when V changes at $T = C$.

$$\left(\frac{\partial U}{\partial p}\right)_T \left(\frac{\partial p}{\partial V}\right)_T \left(\frac{\partial V}{\partial U}\right)_T = 1$$

$$\therefore \quad \left(\frac{\partial U}{\partial p}\right)_T \left(\frac{\partial p}{\partial V}\right)_T = \left(\frac{\partial U}{\partial V}\right)_T = 0$$

Since $\quad \left(\frac{\partial p}{\partial V}\right)_T \neq 0, \left(\frac{\partial U}{\partial p}\right)_T = 0$

U does not change either when p changes at $T = C$. So the internal energy of an ideal gas is a function of temperature only, as shown earlier in Chapter 10. Another important point to note is that in Eq. (11.13), for an ideal gas

$$pV = n\bar{R}T \text{ and } T\left(\frac{\partial p}{\partial T}\right)_v - p = 0$$

Therefore,

$$dU = C_v\, dT$$

holds good for an ideal gas in any process (even when the volume changes). But for any other substance

$$dU = C_v\, dT$$

is true only when the volume is constant and $dV = 0$.
Similarly,

$$dH = TdS + Vdp$$

and

$$TdS = C_p\, dT - T\left(\frac{\partial V}{\partial T}\right)_p dp$$

$$\therefore \quad dH = C_p\, dT + \left[V - T\left(\frac{\partial V}{\partial T}\right)_p\right] dp \tag{11.15}$$

$$\therefore \quad \left(\frac{\partial H}{\partial p}\right)_T = V - T\left(\frac{\partial V}{\partial T}\right)_p \tag{11.16}$$

As shown for internal energy, it can be similarly proved from Eq. (11.16) that the enthalpy of an ideal gas is not a function of either volume or pressure

$$\left[\text{i.e.} \left(\frac{\partial H}{\partial p}\right)_T = 0 \text{ and } \left(\frac{\partial H}{\partial V}\right)_T = 0\right]$$

but a function of temperature alone.
Since for an ideal gas, $pV = n\bar{R}T$

and

$$V - T\left(\frac{\partial V}{\partial T}\right)_p = 0$$

the relation $dH = C_p\, dT$ is true for any process (even when the pressure changes). However, for any other substance the relation $dH = C_p\, dT$ holds good only when the pressure remains constant or $dp = 0$.

2. Thermal radiation in equilibrium with the enclosing walls possesses an energy that depends only on the volume and temperature. The energy density (u), defined as the ratio of energy to volume, is a function of temperature only, or

$$u = \frac{U}{V} = f(T) \text{ only.}$$

The electromagnetic theory of radiation states that radiation is equivalent to a photon gas and it exerts a pressure, and that the pressure exerted by the black-body radiation in an enclosure is given by

$$p = \frac{u}{3}$$

Black-body radiation is thus specified by the pressure, volume and temperature of the radiation.

Since $$U = uV \quad \text{and} \quad p = \frac{u}{3}$$

$$\left(\frac{\partial U}{\partial V}\right)_T = u \text{ and } \left(\frac{\partial p}{\partial T}\right)_V = \frac{1}{3}\frac{du}{dT}$$

By substituting in the energy Eq. (11.13)

$$u = \frac{T}{3}\frac{du}{dT} - \frac{u}{3}$$

$$\therefore \quad \frac{du}{u} = 4\frac{dT}{T}$$

or $$\ln u = \ln T^4 + \ln b$$

or $$u = bT^4$$

where b is a constant. This is known as the *Stefan–Boltzmann Law*.

Since $$U = uV = VbT^4$$

$$\left(\frac{\partial U}{\partial T}\right)_V = C_v = 4VbT^3$$

and $$\left(\frac{\partial p}{\partial T}\right)_V = \frac{1}{3}\frac{du}{dT} = \frac{4}{3}bT^3$$

From the first TdS equation

$$TdS = C_v\, dT + T\left(\frac{\partial p}{\partial T}\right)_V dV$$

$$= 4VbT^3\, dT + \frac{4}{3}bT^4 \cdot dV$$

For a reversible isothermal change of volume, the heat to be supplied reversibly to keep temperature constant

$$Q = \frac{4}{3}bT^4\,\Delta V$$

For a reversible adiabatic change of volume

$$\frac{4}{3}bT^4\, dV = -4VbT^3\, dT$$

$$\therefore \quad \frac{dV}{V} = -3\frac{dT}{T}$$

or $$VT^3 = \text{const.}$$

If the temperature is one-half the original temperature, the volume of black-body radiation is to be increased adiabatically eight times its original volume so that the radiation remains in equilibrium with matter at that temperature.

11.7 JOULE–KELVIN EFFECT

LO 11.6 Explain Joule-Kelvin effect and the significance of inversion curve

A gas is made to undergo continuous throttling process by a valve, as shown in Fig. 11.2. The pressures and temperatures of the gas in the insulated pipe upstream and downstream of the valve are measured with suitable manometers and thermometers.

Let p_i and T_i be the arbitrarily chosen pressure and temperature before throttling and let them be kept constant. By operating the valve manually, the gas is throttled successively to different pressures and temperatures p_{f1}, T_{f1}; p_{f2}, T_{f2}; p_{f3}, T_{f3} and so on. These are then plotted on the *T-p* coordinates as shown in Fig. 11.3. All the points represent equilibrium states of some constant mass of gas, say, 1 kg, at which the gas has the *same enthalpy*.

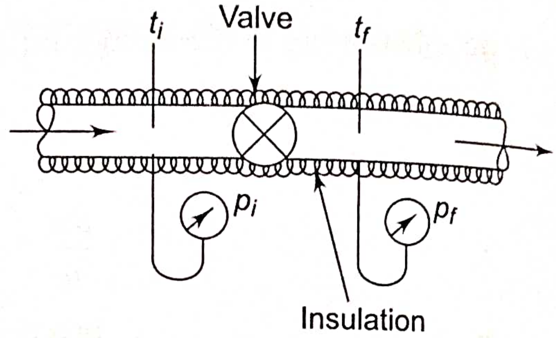

Fig. 11.2 *Joule–Thomson expansion*

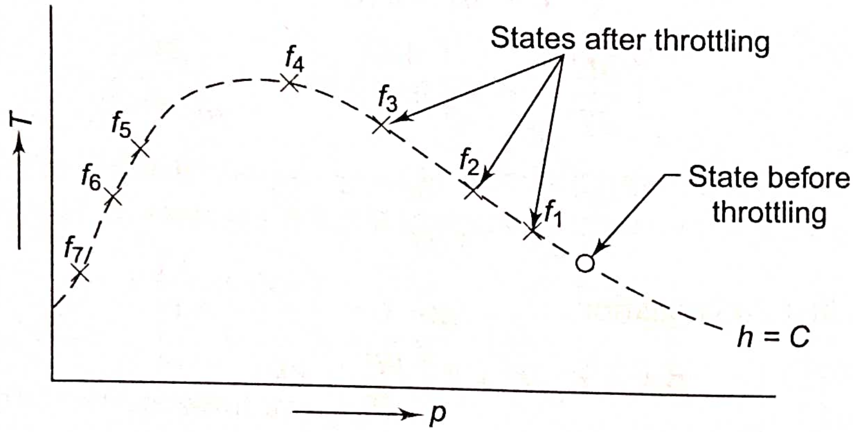

Fig. 11.3 *Isenthalpic states of a gas*

The curve passing through all these points is an isenthalpic curve or an *isenthalpe.* It is not the graph of a throttling process, but the graph through points of equal enthalpy.

The initial temperature and pressure of the gas (before throttling) are then set to new values, and by throttling to different states, a *family of isenthalpes* is obtained for the gas, as shown in Figs. 11.4 and 11.5. The curve passing through the maxima of these isenthalpes is called the *inversion curve.*

The numerical value of the slope of an isenthalpe on a *T-p* diagram at any point is called the *Joule–Kelvin coefficient* and is denoted by μ_J. Thus the locus of all points at which μ_J is zero is the inversion curve. The region inside the inversion curve where μ_J is positive is called the *cooling region* and the region outside where μ_J is negative is called the *heating region.* If a gas is throttled in the cooling region, it will get cooled. If it is throttled in the heating region, the gas will be heated. So,

$$\mu_J = \left(\frac{\partial T}{\partial p}\right)_h$$

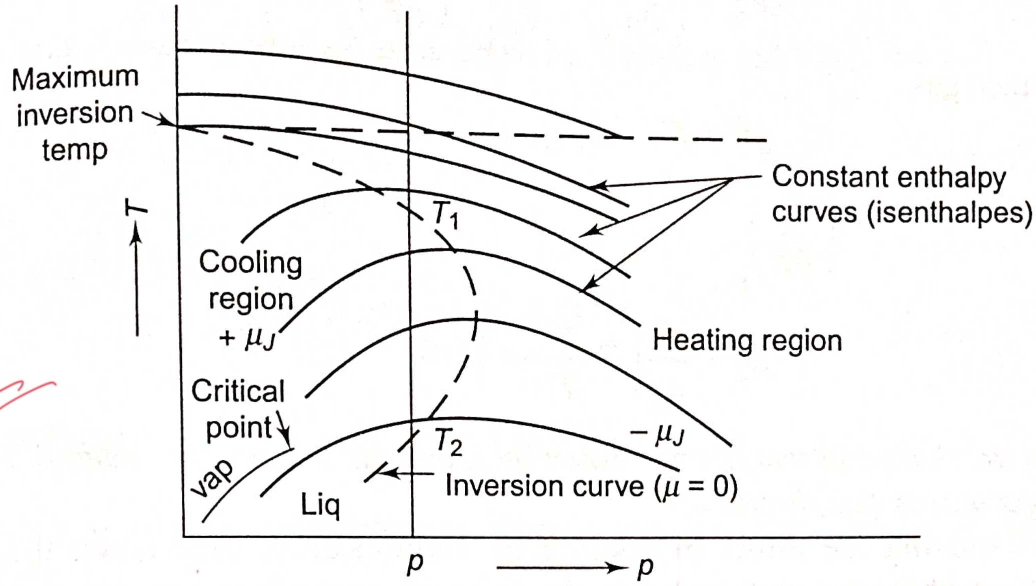

Fig. 11.4 *Isenthalpic curves and the inversion curve*

The difference in enthalpy between two neighbouring equilibrium states is

$$dh = Tds + vdp$$

and the second TdS equation (per unit mass)

$$Tds = c_p\, dT - T\left(\frac{\partial v}{\partial T}\right)_p dp$$

$$\therefore \qquad dh = c_p\, dT - \left[T\left(\frac{\partial v}{\partial T}\right)_p - v\right] dp$$

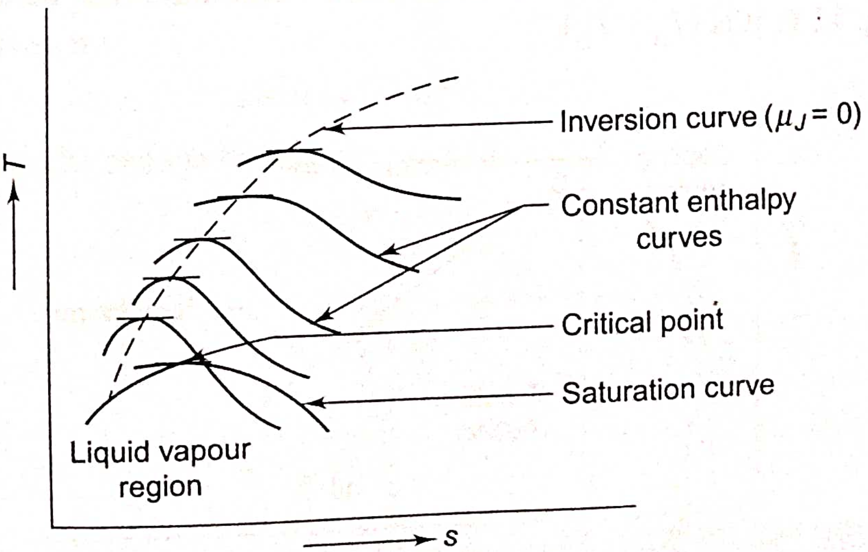

Fig. 11.5 *Inversion and saturation curves on T-s plot*

The second term in the above equation stands only for a real gas, because for an ideal gas, $dh = c_p\, dT$.

$$\mu_J = \left(\frac{\partial T}{\partial p}\right)_h = \frac{1}{c_p}\left[T\left(\frac{\partial v}{\partial T}\right)_p - v\right] \tag{11.17}$$

For an ideal gas

$$pv = RT$$

$$\therefore \quad \left(\frac{\partial v}{\partial T}\right)_p = \frac{R}{p} = \frac{v}{T}$$

$$\therefore \quad \mu_J = \frac{1}{c_p}\left(T \cdot \frac{v}{T} - v\right) = 0$$

There is no change in temperature when an ideal gas is made to undergo a Joule–Kelvin expansion (i.e. throttling).

For achieving the effect of cooling by Joule–Kelvin expansion, the initial temperature of the gas must be below the point where the inversion curve intersects the temperature axis, i.e. below the *maximum inversion temperature*. For nearly all substances, the maximum inversion temperature is above the normal ambient temperature, and hence cooling can be obtained by the Joule–Kelvin effect. In the case of hydrogen and helium, however, the gas is to be precooled in heat exchangers below the maximum inversion temperature before it is throttled. For liquefaction, the gas has to be cooled below the critical temperature.

Let the initial state of gas before throttling be at A (Fig. 11.6). The change in temperature may be positive, zero or negative, depending upon the final pressure after throttling. If the final pressure lies between A and B, there will be a rise in temperature or heating effect. If it is at C, there will be no change in temperature. If the final pressure is below p_C, there will be a cooling effect, and if the final pressure is p_D, the temperature drop will be $(T_A - T_D)$.

Maximum temperature drop will occur if the initial state lies on the inversion curve. In Fig. 11.6, it is $(T_B - T_D)$.

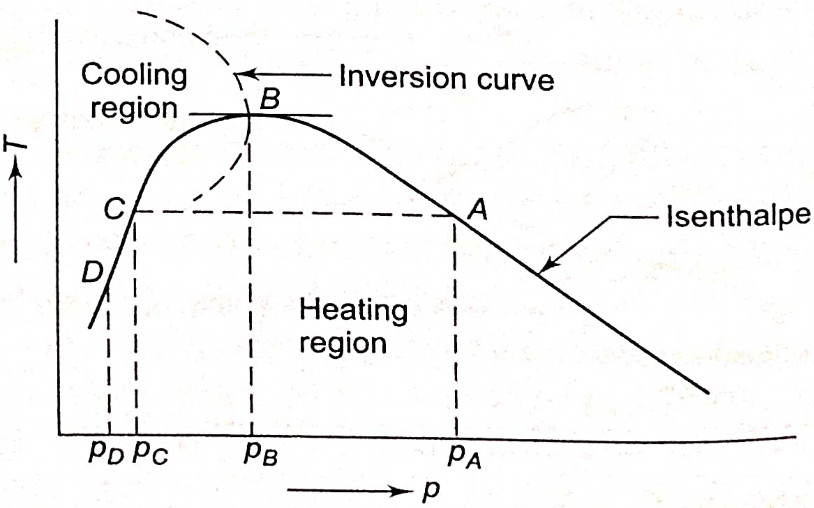

Fig. 11.6 *Maximum cooling by Joule–Kelvin expansion*

The volume expansivity is

$$\beta = \frac{1}{v}\left(\frac{\partial v}{\partial T}\right)_p$$

So the Joule–Kelvin coefficient μ_J is given by, from Eq. (11.17)

$$\mu_J = \frac{1}{c_p}\left[T\left(\frac{\partial v}{\partial T}\right)_p - v\right]$$

or
$$\mu_J = \frac{v}{c_p}[\beta T - 1]$$

For an ideal gas,
$$\beta = \frac{1}{T} \text{ and } \mu_J = 0$$

There are two inversion temperatures for each pressure, e.g. T_1 and T_2 at pressure p (Fig. 11.4).

11.8 CLAUSIUS–CLAPEYRON EQUATION

LO 11.7
Formulate Clausius–Clapeyron equation and discuss how it affects phase change

During phase transitions like melting, vaporisation and sublimation, the temperature and pressure remain constant, while the entropy and volume change. If x is the fraction of initial phase i which has been transformed into final phase f, then

$$s = (1 - x)\, s^{(i)} + x s^{(f)}$$
$$v = (1 - x)\, v^{(i)} + x v^{(f)}$$

where s and v are linear functions of x.

For reversible phase transition, the heat transferred per mole (or per kg) is the latent heat, given by

$$l = T\{s^{(f)} - s^{(i)}\}$$

which indicates the change in entropy.

Now
$$dg = -\, sdT + vdp$$

or
$$s = -\left(\frac{\partial g}{\partial T}\right)_p$$

and
$$v = \left(\frac{\partial g}{\partial p}\right)_T$$

A *phase change of the first order* is known as any phase change that satisfies the following requirements:

1. There are changes of entropy and volume.

2. The first-order derivatives of Gibbs function change discontinuously (Fig. 11.7a).

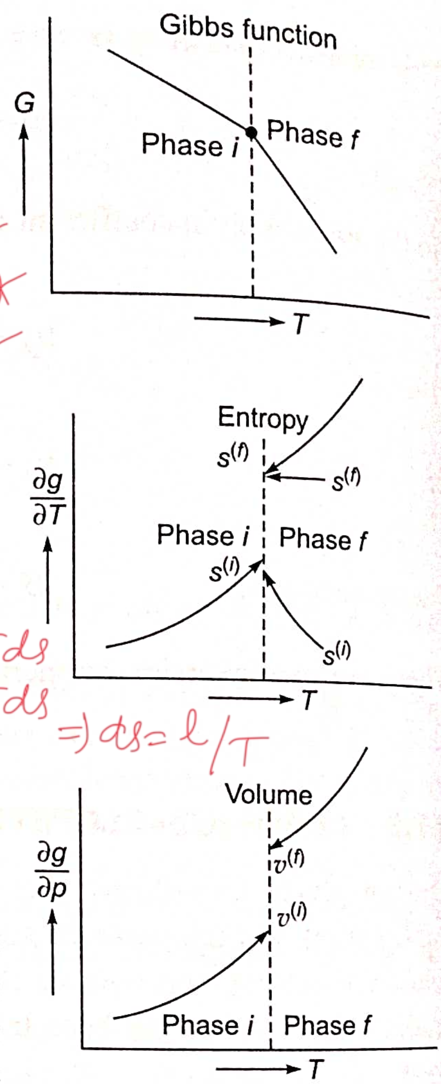

Fig. 11.7(a) *First order phase transition*

Let us consider the first-order phase transition of one mole of a substance from phase i to phase f. Using the first TdS equation

$$Tds = c_v\, dT + T\left(\frac{\partial p}{\partial T}\right)_V dv$$

l → latent heat

for the phase transition which is reversible, isothermal and isobaric, and integrating over the whole change of phase, and since $\left(\frac{\partial p}{\partial T}\right)_V$ is independent of v

$$T\{s^{(f)} - s^{(i)}\} = T\frac{dp}{dT}\{v^{(f)} - v^{(i)}\}$$

dq = Tds

l = Tds

⇒ ds = l/T

$$\therefore \quad \boxed{\frac{dp}{dT} = \frac{s^{(f)} - s^{(i)}}{v^{(f)} - v^{(i)}} = \frac{l}{T\{v^{(f)} - v^{(i)}\}}} \tag{11.18}$$

The above equation is known as the *Clausius–Clapeyron equation.*

The Clausius-Clapeyron equation can also be derived in another way. For a reversible process at constant T and p, the Gibbs function remains constant. Therefore, for the first-order phase change at T and p

$$g^{(i)} = g^{(f)}$$

and for a phase change at $T + dT$ and $p + dp$ (Fig. 11.7b)

$$g^{(i)} + dg^{(i)} = g^{(f)} + dg^{(f)}$$

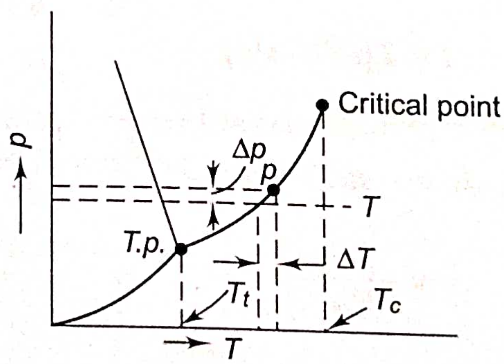

Fig. 11.7(b) *Phase diagram in p-T coordinates where a solid shrinks on melting*

Subtracting

$$dg^{(i)} = dg^{(f)}$$

or

$$-s^{(i)}\, dT + v^{(i)}\, dp = -s^{(f)}\, dT + v^{(f)}\, dp$$

$$\therefore \qquad \frac{dp}{dT} = \frac{s^{(f)} - s^{(i)}}{v^{(f)} - v^{(i)}} = \frac{l}{T\,(v^{(f)} - v^{(i)})}$$

For *fusion*

$$\boxed{\frac{dp}{dT} = \frac{l_{fu}}{T\,(v'' - v')}}$$

where l_{fu} is the latent heat of fusion, the first prime in v, i.e. v' indicates the saturated solid state, and the second prime the saturated liquid state. The slope of the fusion curve is determined by $(v'' - v')$, since l_{fu} and T are positive. If the substance expands on melting, $v'' > v'$, the slope is positive. This is the usual case. Water, however, contracts on melting and has the fusion curve with a negative slope (Fig. 11.8).

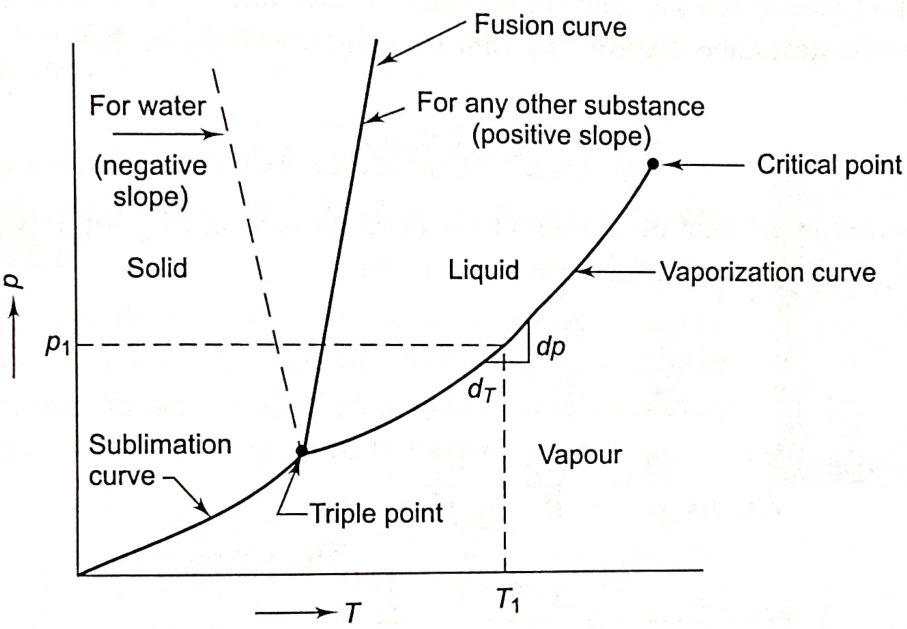

Fig. 11.8 *Phase diagram for water and any other substance on p-T coordinates*

For *vaporisation*

$$\boxed{\frac{dp}{dT} = \frac{l_{vap}}{T\,(v''' - v'')}}$$

where l_{vap} is the latent heat of vaporisation, and the third prime indicates the saturated vapour state.

$$\therefore \qquad l_{vap} = T\,\frac{dp}{dT}\,(v''' - v'')$$

At temperature considerably below the critical temperature, $v''' >> v''$ and using the ideal gas equation of state for vapour

$$v''' = \frac{RT}{p}$$

$$l_{vap} \cong T \cdot \frac{dp}{dT}\,\frac{RT}{p}$$

or

$$\boxed{l_{vap} = \frac{RT^2}{p}\,\frac{dp}{dT}} \qquad (11.19)$$

If the slope dp/dT at any state (e.g. point p_1, T_1 in Fig. 11.8) is known, the latent heat of vapourisation can be computed from the above equation.

The vapour pressure curve is of the form

$$\ln p = A + \frac{B}{T} + C \ln T + DT$$

where A, B, C and D are constants. By differentiating with respect to T

$$\frac{1}{p}\frac{dp}{dT} = -\frac{B}{T^2} + \frac{C}{T} + D \tag{11.20}$$

Equations (11.19) and (11.20) can be used to estimate the latent heat of vaporisation.

Clapeyron's equation can also be used to estimate approximately the vapour pressure of a liquid at any arbitrary temperature in conjunction with a relation for the latent heat of a substance, known as *Trouton's rule*, which states that

$$\frac{\bar{h}_{fg}}{T_B} \cong 88 \text{ kJ/kg mol K}$$

where h_{fg} is the latent heat of vapourisation in kJ/kg mol and T_B is the boiling point at 1.013 bar. On substituting this into Eq. (11.19)

$$\frac{dp}{dT} = \frac{88T_B}{\bar{R}T^2}\, p$$

or

$$\int_{101.325}^{p} \frac{dp}{p} = \frac{88T_B}{\bar{R}} \int_{T_B}^{T} \frac{dT}{T^2}$$

$$\ln \frac{p}{101.325} = -\frac{88T_B}{\bar{R}}\left(\frac{1}{T} - \frac{1}{T_B}\right)$$

$$\therefore \qquad p = 101.325 \exp\left[\frac{88}{R}\left(1 - \frac{T_B}{T}\right)\right] \tag{11.21}$$

This gives the vapour pressure p in kPa at any temperature T.

For *sublimation*

$$\frac{dp}{dT} = \frac{l_{sub}}{T(v''' - v')}$$

where l_{sub} is the latent heat of sublimation.

Since $v''' >> v'$, and vapour pressure is low, $v''' = \frac{RT}{p}$

$$\frac{dp}{dT} = \frac{l_{sub}}{T \cdot \frac{RT}{p}}$$

or

$$l_{sub} = -2.303\, R \frac{d(\log p)}{d(1/T)}$$

the slope of log p *vs.* $1/T$ curve is negative, and if it is known, l_{sub} can be estimated. At the triple point (Fig. 9.12),

$$l_{sub} = l_{vap} + l_{fus} \tag{11.23}$$

$$\left(\frac{dp}{dT}\right)_{vap} = \frac{p_{tp} l_{vap}}{\bar{R} T_{tp}^2}$$

$$\left(\frac{dp}{dT}\right)_{sub} = \frac{p_{tp} l_{sub}}{\bar{R} T_{tp}^2}$$

Since $l_{sub} > l_{vap}$, at the triple point

$$\left(\frac{dp}{dT}\right)_{sub} > \left(\frac{dp}{dT}\right)_{vap}$$

Therefore, the slope of the sublimation curve at the triple point is greater than that of the vapourisation curve (Fig. 11.8).